

The Elephant Random Walk Across Its Full Phase Diagram: Exact Moments, Boundary Martingales, and Limit Laws

HCS-C316 theorem-and-evidence package

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Abstract

We give a boundary-complete account of the one-dimensional elephant random walk for every memory parameter $p \in [0, 1]$ and initial bias $q \in [0, 1]$. Exact recurrences yield the first two moments; the singular cases $p = 0$ and $p = 3/4$ receive separate martingale and harmonic formulas. The diffusive, critical, and superdiffusive limits are stated with exact constants, and the first four moments of the superdiffusive limit expose the deterministic and two-point endpoints at $p = 1$. A content-addressed exact computation audits finite identities but is explicitly not offered as proof of an asymptotic theorem.

1 Model and finite laws

Let $X_1 \in \{-1, 1\}$ satisfy $\Pr(X_1 = 1) = q$. Given $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, choose K uniformly in $\{1, \dots, n\}$ and set $X_{n+1} = X_K$ with probability p and $X_{n+1} = -X_K$ otherwise. Put

$$S_n = \sum_{j=1}^n X_j, \quad a = 2p - 1, \quad b = 2q - 1, \quad G_n(c) = \prod_{j=1}^{n-1} \left(1 + \frac{c}{j}\right).$$

The model originates with Schütz and Trimper [1]; the martingale phase analysis used below follows Bercu [2].

Theorem 1 (Exact kernel and moments). *For every $n \geq 1$,*

$$\Pr(X_{n+1} = 1 \mid \mathcal{F}_n) = \frac{1}{2} \left(1 + \frac{aS_n}{n}\right), \quad \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = \frac{aS_n}{n}, \quad (1)$$

$$\mathbb{E}S_n = bG_n(a). \quad (2)$$

Furthermore, with $H_n = \sum_{j=1}^n j^{-1}$,

$$\mathbb{E}S_n^2 = \begin{cases} \frac{2aG_n(2a) - n}{2a - 1}, & a \neq \frac{1}{2}, \\ nH_n, & a = \frac{1}{2}. \end{cases} \quad (3)$$

If $p > 0$, $M_n = S_n/G_n(a)$ is a martingale. At $p = 0$ instead, $S_2 = 0$ and $\tilde{M}_n = (n-1)S_n$, $n \geq 2$, is a martingale.

Proof. Averaging the signed copy over K proves (1). Hence

$$\mathbb{E}(S_{n+1} \mid \mathcal{F}_n) = \left(1 + \frac{a}{n}\right) S_n, \quad \mathbb{E}(S_{n+1}^2 \mid \mathcal{F}_n) = \left(1 + \frac{2a}{n}\right) S_n^2 + 1.$$

Iteration gives (2); variation of constants gives (3), with the resonant sum equal to H_n when $2a = 1$. For $p > 0$, divide the first recurrence by $G_{n+1}(a) = (1 + a/n)G_n(a)$. At $p = 0$, this product vanishes after its first factor, but direct substitution of $a = -1$ gives $\mathbb{E}[nS_{n+1} \mid \mathcal{F}_n] = (n-1)S_n$; also $X_2 = -X_1$. $\square$

Revision certificate: exact finite structure

This original round contains the exact kernel, both moment branches, and the repaired $p = 0$ martingale.

References

- [1] G. M. Schütz and S. Trimper, “Elephants can always remember,” *Phys. Rev. E* 70 (2004), DOI: 10.1103/PhysRevE.70.045101.
- [2] B. Bercu, “A martingale approach for the elephant random walk,” *J. Phys. A* 51 (2018), DOI: 10.1088/1751-8121/aa95a6.